

Resonance-Based Subspace Dynamics as a Unifying Framework for Particles and Fundamental Interactions

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Abstract

We present a resonance-based subspace model in which particles and fundamental interactions emerge as stable eigenmodes of a structured, continuous subspace. In contrast to point-particle descriptions, matter is modeled as localized standing wave configurations arising from coupled orthogonal modes. Interactions result from interference, phase rotation, high-frequency mode coupling, and mode transitions within the same underlying medium.

Gravitation is described as a geometric deformation induced by the resonance energy of the subspace, electromagnetic phenomena as internal phase rotations of multimodal structures, the strong interaction as short-range high-frequency coupling, and the weak interaction as unstable mode transitions. Quantization follows naturally from resonance stability conditions, while relativistic and quantum field equations appear as limiting cases of the subspace field dynamics. The approach provides a physical interpretation of mass, charge, and particle stability and leads to testable predictions, including resonance-dependent gravitational effects, phase-dependent photon propagation, and energy shifts of nucleonic states under extreme rotational or pressure conditions. Cosmological implications include galactic rotation profiles without invoking dark matter, interpreted as large-scale vortex structures of the subspace.

This work is intended as a unifying, mechanically interpretable extension of existing quantum mechanical and relativistic theories, based on resonance dynamics and, in principle, experimentally accessible predictions.